

Khashi Ghorbani

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Education	Ph.D., Agricultural and Applied Economics, University of Illinois Urbana-Champaign, USA M.Sc. in Environmental Economics, University of New Hampshire, USA M.Sc. in Agricultural Economics, University of Tehran, Iran B.Sc. in Industrial Engineering, Karaj Azad University, Iran	Expected 2026 2021 2017 2013
Research Areas	Agricultural Economics, Resource Economics, Bioeconomic	
Publications	Ghorbani, K., Atallah, S., Gallardo, K., (2024) "Commentary on Extension Programming: An Online Platform for Area-Wide Management of Western X-Disease Extension Program" ., Applied Economics Teaching Resources (AETR). (Accepted) "This paper presents an extension program developed using a logic model to assist in the area-wide management of Western X-disease through an online decision-aid tool. The program provides research-based recommendations on optimal tree removal, fosters grower commitment, and highlights the economic benefits of coordinated disease management." Ghorbani, K., & Mashange, G. (2024). The Dynamics of Sweet Cherry Orchards: Trends and Transitions from 2008 to 2023. farmdoc daily, 14(151). <i>"This study examines the evolution of U.S. sweet cherry production, highlighting Washington's dominance, California's expansion, and Oregon's decline due to environmental constraints. It also explores land use transitions and the economic impact of pest and disease pressures, emphasizing the industry's need for adaptive strategies."</i>	
In review	Ghorbani, K., Atallah, S. (2024) "Climate-change impacts mediated by crop diseases: The case of coffee leaf rust" ., American Journal of Agricultural Economics (AJAE). (Revision and resubmit) "This study quantifies the direct and indirect effects of climate change on coffee production, emphasizing the role of coffee leaf rust in shaping economic outcomes. Using a bioeconomic model, it highlights the differential adaptation priorities for coffee farms at high and low altitudes and identifies climate-adjusted price premiums to sustain shade-grown coffee systems." Ghorbani, K., Atallah, S. (2024) "Strategic Responses to Ban Enforcement Uncertainty: Antibiotic Application Decisions in Plant Agriculture" ., European Review of Agricultural Economics (ERAЕ). (Revision and resubmit) "This study develops a dynamic model to examine optimal antibiotic use in plant agriculture under uncertain ban enforcement, revealing that proactive growers overexploit antibiotic efficacy to gain an advantage over business-as-usual growers. The analysis highlights the nonlinear relationship between resource value and regulatory uncertainty, proposing a policy instrument to compensate proactive growers if anticipated bans do not materialize."	
Working Papers and Research in Progress	Ghorbani, K., Atallah, S., Gallardo, K., (2024) "Growers' Willingness to Adopt Non-Antibiotic Disease Management Strategies to Cope with Antibiotic Resistance" "This study employs a discrete choice experiment to assess apple growers' preferences for non-antibiotic disease management strategies amid rising concerns over antibiotic resistance. Findings reveal that long-term effectiveness, peer adoption, and information dissemination significantly influence growers' willingness to invest in sustainable alternatives, highlighting the need for targeted incentives and collective action in resistance management." Ghorbani, K., Atallah, S., Gallardo, K., (2024) "From Detection to Recovery: An Economic Perspective on Managing Infectious Disease in Orchards" "This study develops a spatial bioeconomic model to evaluate optimal tree removal and replanting strategies for managing Western X-Disease in cherry orchards. This paper discusses under what condition, proactive	

disease management, including testing and strategic replanting, mitigates economic losses and enhances long-term orchard profitability.”

Ghorbani, K., Atallah, S., Gallardo, K., Tianna Dupont., (2024) “Return on Investment in Alternative Fire Blight Management Strategies: Evidence from Field Trials in Washington State”

“This study assesses the economic viability of various biopesticide and systemic acquired resistance (SAR) treatments for fire blight management using return on investment (ROI) analysis. Findings highlight that treatment efficacy and economic feasibility depend on disease severity, with some alternatives providing significant financial benefits under high infection pressure.”

Winfree, J., Atallah, S., Gallardo, K., **Ghorbani, K.** (2024) “The Economic Effects of Persistent Plant Diseases: The Case of Fire blight”

“This study examines the economic impact of fire blight on pear production, differentiating between anticipated and unanticipated disease shocks. Using a Cournot model and empirical price-quantity relationships, the analysis reveals that fire blight reduces industry yield and welfare, with mixed effects on producer profits depending on disease severity and market conditions.”

Ghorbani, K. (2024) “A Game-Theoretic Mechanism to Combat Antibiotic Resistance in Livestock Production”

“This study develops a game-theoretic compensation mechanism to incentivize farmers to reduce antibiotic use while internalizing resistance-related externalities. Findings highlight the role of farmers' perceptions in the mechanism's effectiveness and suggest that aligning compensation strategies with education campaigns enhances antibiotic stewardship in agriculture.”

Ghorbani, K., Jared Hutchins, Jonathan Coppess (2025) “Spatial Distribution of Farm Service Agency (FSA) Farm Payments”

“We are curating FSA payment dataset to enhance data availability for researchers, policymakers, and agricultural stakeholders through the National Center for Supercomputing Applications (NCSA).”

Relevant Work Experience

Research Assistant, University of Illinois Urbana-Champaign, USA Fall 2021 – present
Under Dr. Shadi Atallah contributed to two projects. In the comprehensive fire blight management project (funded by USDA-NIFA and in collaboration with Michigan State University, Washington State University, and Cornell University), we analyze the economic impact of using non-antibiotic treatments and find growers' willingness to pay for non-antibiotic treatments. We survey growers from top apple producing states including Washington, Michigan, New York, California, Oregon, Pennsylvania, Wisconsin, and Ohio. The second project is National Clean Product Network Economic Research (funded by USDA-APHIS and in collaboration with Washington State University) and we analyze potential economic gains of area-wide management for individual growers and develop a bioeconomic decision model to find the optimal threshold and intensity of diseased tree removal.

Research/Teaching Assistant, University of New Hampshire, USA Fall 2019 – Summer 2021
Teaching assistant to Prof. Alberto Manalo in Environmental Economics course - Undergraduate level. 70-90 students signed up for this course each semester. Also, under Dr. Shadi Atallah developed economic analysis of aquaponic systems in New Hampshire and provided economic returns of coupled and decoupled systems of producing Tilapia and lettuce.

Research Associate, University of Aveiro, Portugal Summer 2018 - Spring 2019
The project was to analyze the economic drivers of food security in rural areas of Portugal (funded by Department of Economics at University of Aveiro). My contribution was to prepare the survey instrument and run the expert group interviews.

Economist, Macro-Economic Research Institute (MAKAN), Iran Summer 2016 - Spring 2018
Two main projects in the research unit at MAKAN. One, we developed Food Security Atlas in Khuzestan Province, Iran. I collaborated with health and nutrition specialists to find the economic drivers of food security and analyze the impact on HFIAS food security score. The other project was a comprehensive study on Wheat guaranteed price policy in Iran. We determined that the actual cost of this policy is nearly four times higher than the allocated budget and recommended cost saving resolutions along supply chain.

	Research/Teaching Assistant, University of Tehran, Iran	Fall 2014 – Spring 2016
	Teaching assistant to Dr. Saeed Yazdani in the Sustainable Development course - Undergraduate and master student level. 10-15 students signed up for this course each semester. Also, under Dr. Hamed Rafiee contributed to the economic analysis of water market in Iran. Under a simulated scenario, we found that the industry section would take control of over 90% of water use permits given the proximity of this section to underground water resources.	
	Business Analyst, Sarmye-saze-qarb Business consulting firm, Iran	Fall 2012 – Summer 2014
	As an intern and business analyst in the business analytics unit, I developed economic feasibility studies for food industry clients.	
Teaching	Environmental Economics, University of New Hampshire Sustainable Development, University of Tehran ALOHA software instructor, Karaj Azad University	Fall 2019 – Spring 2021 Fall 2014 – Winter 2015 Fall 2012 – Winter 2013
Fellowships, Grants and Academic Awards	Outstanding PhD student award, Department of Agricultural and Consumer Economics, University of Illinois Urbana-Champaign 1 st place, Graduate student extension competition, Agricultural and Applied Economics Association Annual Meeting Travel Grant, University of Illinois Urbana-Champaign Travel Grant, University of Illinois Urbana-Champaign Travel Grant, University of Illinois Urbana-Champaign Travel Grant, University of Illinois Urbana-Champaign	2025 2024 Fall 2024 Summer 2024 Spring 2024 Spring 2022
Presentations	Agricultural & Applied Economics Association, Denver, USA (Planned) Association of Environmental and Resource Economists, Santa Ana Pueblo, USA (Planned) 25 th BIOECON annual conference, Cambridge, UK Agricultural & Applied Economics Association, New Orleans, USA World Conference on Natural Resource Modeling, New Haven, USA The fire blight management annual symposium, Traverse city, USA Conference on degrowth and sustainability, West Lafayette, USA The Agricultural & Applied Economics Association, Virtual The United States Society for Ecological Economics Conference, Virtual The 11 th biennial conference of Iran's Agricultural Economics, Karaj, Iran The first International Conference of Engineering and Agriculture, Environment and Sustainable Natural Resources, Hamedan, Iran The 10th biennial conference of Iran's Agricultural Economics, Kerman, Iran	2025 2025 2024 2024 2024 2024 2021 2021 2018 2016 2016
Service and Leadership	<i>Reviewer of Applied Economics Letters</i> <i>Graduate Student Organization Officer</i> , Department of Agricultural and Consumer Economics, University of Illinois Urbana-Champaign <i>Vice President of Agricultural Economics Student Association</i> , Department of Agricultural Economics, University of Tehran	2025 2024-2025 2016-2017
Additional Information	Software and Programing: Python, Stata, R, JavaScript, Microsoft Office Languages: English, Farsi (Native), Spanish (Basic)	